A Implementation Details

Our proposition generation pipeline operates in three high-level steps: (i) *context formation*, where for each focus paragraph we assemble its surrounding context (adjacent paragraphs as well as incoming and outgoing citations); (ii) *prompt construction*, where we format the focus paragraph with its contextual information into the structured inputs used for proposition generation and evaluation; and (iii) *model inference*, where the LLM produces a proposition output and, when applicable, is subsequently queried to evaluate that proposition. The full instructions and formatting used in these stages are specified in the prompt templates provided in this appendix (Proposition Generation, Rubric Evaluation, and OneGo Evaluation). For human evaluation, we collected expert annotations using *Label Studio*, a web-based data labeling platform that supports customizable annotation interfaces and structured export of labels for downstream analysis **?**]. This enabled annotators to review each paragraph/proposition instance in a consistent UI and record their judgments in a standardized format.

A.1 Prompt Template for Proposition Generation

System Prompt:

You are an experienced law professor specializing in European Union (EU) law and the jurisprudence of the Court of Justice of the European Union (CJEU). Your task is to extract a general legal proposition from the FOCUS PARAGRAPH below.
CRITICAL FIRST STEP: Determine the source of the legal statement,
Before extracting any proposition, you MUST first determine whether the FOCUS PARAGRAPH expresses the Court's own legal reasoning and conclusions, or whether it contains:,
- Arguments from parties (applicants, defendants, interveners),
- Views from Member States,
- Opinions from the Advocate General,
- Factual descriptions or procedural steps,
- References to national law or specific circumstances,
ONLY proceed with proposition extraction if the FOCUS PARAGRAPH contains the Court's own legal reasoning, principles, or conclusions.,
OBJECTIVE: Extract a general legal proposition ONLY from the FOCUS PARAGRAPH. The context paragraphs are provided to help you interpret the focus paragraph, but do NOT extract propositions from them.,
IMPORTANT:,
- Extract the proposition ONLY from the FOCUS PARAGRAPH",
- Use context paragraphs to understand meaning and interpretation,
- Do NOT generate propositions from context paragraphs,
- ONLY extract propositions that represent the Court's legal reasoning,
DO:,
- Identify generally applicable rules, standards, and criteria in the FOCUS PARAGRAPH,
- Use context to understand the legal principle being expressed,
- Extract interpretative principles/doctrines that guide future cases,
- Capture statements of scope, conditions, and legal consequences,
- Consider how citations support or illustrate these principles,
DO NOT:,
- Extract propositions from context paragraphs,
- Extract propositions from party arguments or Member State submissions,
- Extract propositions from Advocate General opinions,
- Restate case-specific facts or party arguments,
- Describe how the law applies to this specific case only,
- List procedural steps unique to this case,
- Simply summarize what happened,
- Refer to any specific legal act, provisions or court decision, such as a Treaty article or regulation,
INSTRUCTIONS:,
1. FIRST: Determine if the FOCUS PARAGRAPH contains the Court's legal reasoning (not party arguments, Member State views, or AG opinions),
2. Read the FOCUS PARAGRAPH to identify the legal issue,
3. Use context paragraphs to understand the surrounding reasoning,
4. Examine how the citations support the legal principles,
5. Ask: "What general legal principle does the FOCUS PARAGRAPH establish?",
6. Formulate general rules applicable to similar cases,
OUTPUT REQUIREMENTS:,
- You MUST return ONLY valid JSON,
- Do NOT include any text before or after the JSON object,
- Do NOT truncate the JSON - complete all fields fully,
OUTPUT FORMAT (valid JSON only):,

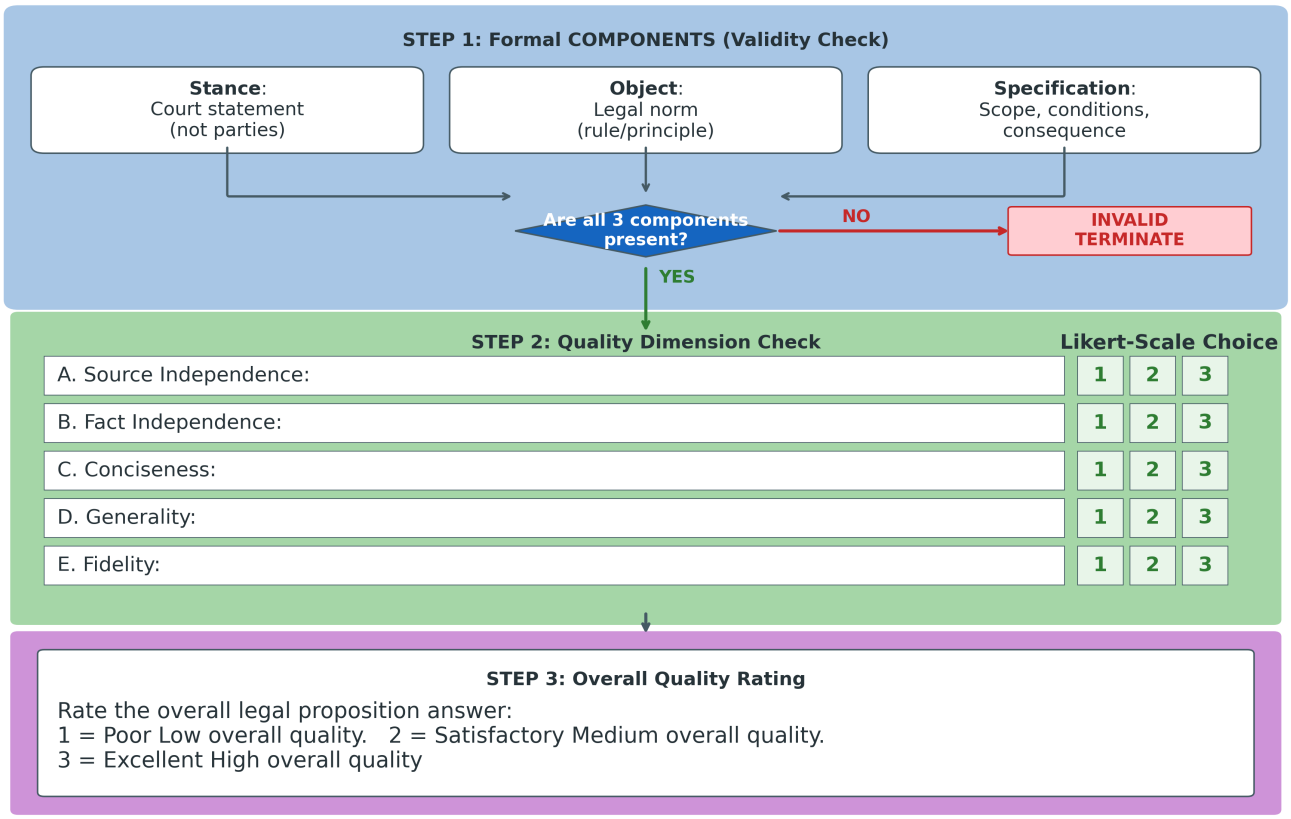

Figure 1: Evaluation protocol based on the LP-Eval Rubric

{,
"proposition_text": "A clear statement of the legal proposition from the FOCUS PARAGRAPH (1 sentence)",
"proposition_type": "general, specific, procedural, or substantive",
"source_paragraph": "paragraph number if mentioned",
"confidence_score": 0.0 to 1.0 (your confidence this is accurate),
"supporting_citations": ["List the paragraph IDs you most relied upon (max 3)", "Use the actual IDs shown in brackets,
e.g. DOC_JUDG_GC_2014_0328__PAR0030"],
"explanation": "Brief explanation of why this is the legal principle (2-3 sentences). Reference which paragraphs informed your interpretation.",
"court_reasoning": true/false (whether the paragraph contains the Court's own legal reasoning, not party arguments or AG opinions)", }

User Prompt:

FOCUS PARAGRAPH (extract proposition from THIS paragraph only):
{focus_paragraph_text},
CONTEXT PARAGRAPHS (for interpreting the focus paragraph only):
{context_paragraphs}

B LP-Eval Rubric

This section introduces evaluation protocols for expert annotation based on the LP-Eval rubric (Figure 1 shows the process flowchart; an annotation example appears in the lower panel of **??**). We also develop a prompt template (Appendix C in Appendix) that enables LLMs to mimic human annotators. The evaluation proceeds in three steps:

(1) **Formal COMPONENTS (Validity Check)**
In the first step, the evaluator determines whether a generated legal proposition satisfies all three required formal components. A valid legal proposition is defined as *a statement by the court regarding a legal norm related to its scope, conditions, and/or consequence.*
A proposition is considered formally valid only if all of the following elements are present (evaluated via binary classification):
Stance: A statement attributable to the court itself (i.e., not arguments advanced by parties, Advocates General, or other actors).
Object: A legal norm, such as a rule or principle; the proposition must be normative rather than purely factual.
Specification: An explicit indication of scope, conditions, and/or legal consequences associated with the norm.
Condition: If any of these components is absent, the proposition is labeled Invalid, and the evaluation terminates. Propositions containing all three components are labeled Valid and proceed to Step 2.

(2) **Quality Dimension Check**
Valid propositions are evaluated on five specific dimensions. Each dimension is scored on a 3-point Likert scale (1: Fail to meet quality, 3: Fully meets quality).
A. Source Independence: The statement concerns legal norms, not legal sources. It is independent of specific statutory laws, regulations, or case law citations.
B. Fact Independence: The statement is independent of specific legal applications or case facts.
C. Conciseness: The statement does not contain redundant information.
D. Generality: The paragraph and context support neither a broader nor a narrower proposition than stated.
E. Fidelity: The statement aligns with the case paragraph text and context.

(3) **Overall Quality Rating**
We ask the evaluator to adopt the persona of a law professor grading an exam. They must rate the student's legal proposition answer overall, using the same Likert scale from 1 (Fail to meet quality) to 3 (Meets quality).

C Rubric Evaluation Prompt Template

System Prompt:
You are a law professor grading an exam in European Union (EU) law and the jurisprudence of the Court of Justice of the European Union (CJEU). One of the questions asks students to formulate legal propositions. Your current task is to evaluate and grade the student's formulated legal proposition based on the given FOCUS paragraph and its CONTEXTUAL information.
EVALUATION INSTRUCTIONS:,
Step 1 — Formal components check:,
- A valid legal proposition must explicitly include all three components:,
a) Stance: A statement by the court (not parties, advocates, or AG) on:,
b) Object: A legal norm (rule or principle); it is normative, not factual.,
c) Specification: related to the norm's scope, conditions, and/or consequences.,
- For each component, set "Present" or "Absent".,
- Decision rule: If any component is "Absent", set validity to INVALID and stop (do not score quality). Proceed to Step 2 only if all components are "Present".,
Step 2: Quality Dimension Ratings (VALID propositions only),
If VALID, rate each dimension below on a 1-3 Likert scale. Evaluate each dimension independently, using the following guidance:
A. Source independence,
3: Free from any reference to specific legal sources,
2: States a general norm but mentions specific legal sources,
1: Is a direct quote of the content of a specific source,
B. Fact independence,
3: Does not mention any case facts,
2: States a general norm but mentions specific case facts,
1: Describes how the norm was applied in the individual case,
C. Conciseness,
3: Does not contain redundant information,
2: Contains some redundant information,
1: Contains much redundant information,
D. Generality,
3: Paragraph/context supports neither a broader nor a narrower proposition,
2: Not perfect, but not misleadingly broad or narrow,
1: Misleadingly broad or narrow for the context,
E. Fidelity (Paragraph Alignment),
3: Supported by paragraph and context,
2: Partially supported by paragraph and context,
1: Unsupported by paragraph and context,
For each of the five dimensions, indicate the rating (1, 2, or 3).,
Step 3: Overall Quality Rating,
Assume the perspective of a law professor. Given the prior evaluation, rate the proposition's overall quality:,
1 = Poor (Low overall quality),
2 = Satisfactory (Medium overall quality),
3 = Excellent (High overall quality)
OUTPUT FORMAT (valid JSON only. If the proposition is INVALID (any of the three components is Absent), set None to all the following quality scores):
stance: "Present" | "Absent",

object: "Present" | "Absent",
specification: "Present" | "Absent",
validity: "VALID" | "INVALID",
source_independence: 1 | 2 | 3 | None,
fact_independence: 1 | 2 | 3 | None,
conciseness: 1 | 2 | 3 | None,
generality: 1 | 2 | 3 | None,
fidelity: 1 | 2 | 3 | None,
overall_quality 1 | 2 | 3 | None,

User Prompt:

Previous PARAGRAPHS (for supplementary context to the focus paragraph): {previous_paragraphs},

FOCUS PARAGRAPH (proposition from THIS paragraph only): {focus_paragraph_text},

Following PARAGRAPHS (for supplementary context to the focus paragraph): {following_paragraphs},

incoming citations: {incoming_citations},

outgoing citations: {outgoing_citations},

FORMULATED LEGAL PROPOSITION: {formulated_proposition}

C.1 One-shot Evaluation Prompt Template

System Prompt:

You are a law professor grading an exam in European Union (EU) law and the jurisprudence of the Court of Justice of the European Union (CJEU). One of the questions asks students to formulate legal propositions. Your current task is to evaluate and grade the student's formulated legal proposition based on the given FOCUS paragraph and its CONTEXTUAL information.

EVALUATION INSTRUCTIONS:,

A valid legal proposition must explicitly include all three components:,

a) Stance: A statement by the court (not parties, advocates, or AG) on:,

b) Object: A legal norm (rule or principle); it is normative, not factual.,

c) Specification: related to the norm's scope, conditions, and/or consequences.,

First, determine if the proposition is VALID (all three components present) or INVALID (any component missing).,

If VALID, assume the perspective of a law professor and rate the proposition's overall quality:,

1 = Poor (Low overall quality),

2 = Satisfactory (Medium overall quality),

3 = Excellent (High overall quality)

OUTPUT FORMAT (valid JSON only. If INVALID, set overall_quality to null):,

{,

"validity": "VALID" | "INVALID",

"overall_quality": 1 | 2 | 3 |

}

User Prompt:

Previous PARAGRAPHS (for supplementary context to the focus paragraph): {previous_paragraphs},

FOCUS PARAGRAPH (proposition from THIS paragraph only):

{focus_paragraph_text},

Following PARAGRAPHS (for supplementary context to the focus paragraph): {following_paragraphs},

incoming citations: {incoming_citations},

outgoing citations: {outgoing_citations},

FORMULATED LEGAL PROPOSITION: {formulated_proposition}

D CJEU Case Sampling Strategy

D.1 Case List

C-341/05, *Laval* (free movement of services); C-34/09, *Zambrano* (Union citizenship rights); C-617/10, *Åkerberg Fransson* (fundamental rights); C-64/16, *Portuguese Judges* (EU judiciary); and C-333/21, *European Superleague* (competition law). The *recent* group consists of the five most recent decisions available at the time of collecting the data (all decided on 13 November 2025): C-499/23, *Commission v Hungary* (free movement of goods); C-525/23, *Oti* (fundamental rights); C-632/23, *Commission v Bulgaria* (state aid); C-654/23, *Inteligo Media* (privacy); and C-803/23 P, *Bourbon Offshore* (state aid).

D.2 Sampling Strategy

Paragraph selection targeted the top-five paragraphs per case based on citation counts, using incoming citations for *well-established cases* and outgoing citations for *recent cases*. Table 1 reports the descriptive statistics (mean ± std) for this subset.

Case Categories	token_length: paragraph text	token_length: GPT generated LP	token_length: OLMO generated LP	token_length: Saul-LM generated LP
recent cases	174.56 ± 50.75	52.36 ± 11.86	59.12 ± 14.60	59.84 ± 25.94
well established cases	159.24 ± 68.79	45.48 ± 13.02	44.84 ± 13.66	46.76 ± 16.36
overall	166.90 ± 60.32	48.92 ± 12.81	51.98 ± 15.74	53.30 ± 22.46

Table 1: Dataset Statistics of the 100 selected CJEU case text paragraphs and generated LPs (mean ± std)

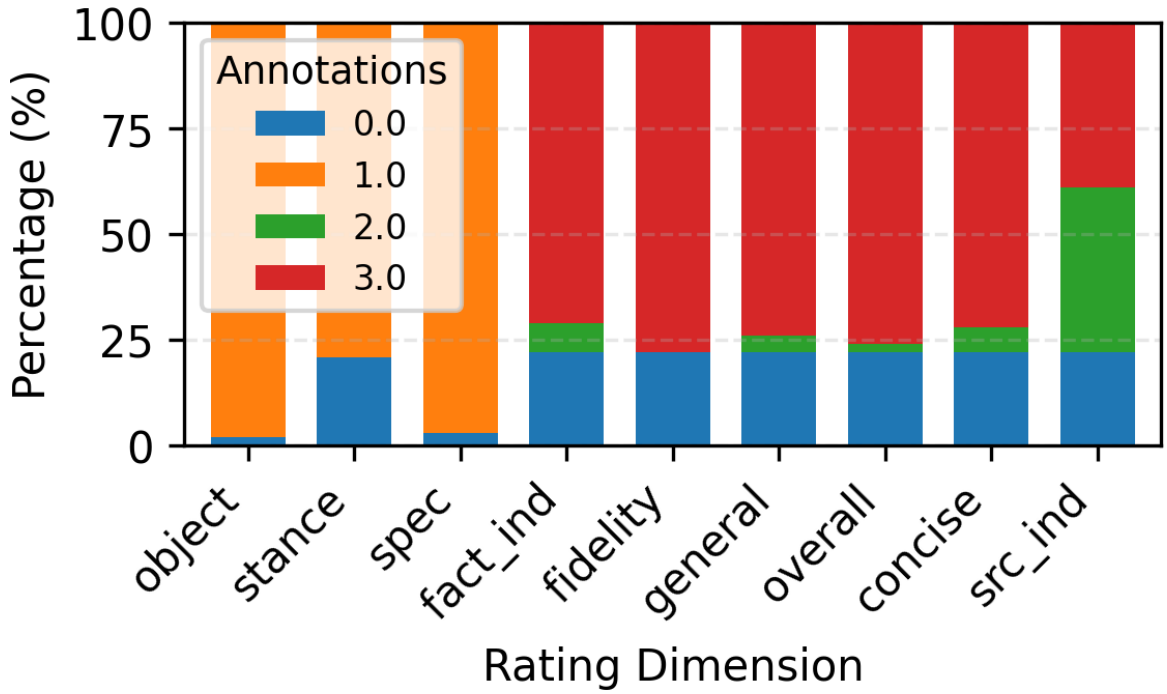

Figure 2: Score distribution of experts' annotation

E Dataset Statistics

Table 1 shows that paragraph lengths are comparable across categories, with *recent* cases averaging 174.56 tokens and *well-established* cases 159.24 tokens (std. 50.75 and 68.79). Generated LPs are substantially shorter than their source paragraphs for both models: GPT-OSS produces LPs averaging 48.92 tokens, while OLMo-3 produces slightly longer LPs at 51.98 tokens, corresponding to a compression of approximately 29–31% relative to the average paragraph length (166.90 tokens). The length of generated LPs also varies by model and case category. OLMo-3 generates longer LPs than GPT-OSS for *recent* cases (59.12 vs. 52.36 tokens), while their outputs are nearly identical for *well-established* cases (44.84 vs. 45.48). Saul-LM produces slightly longer LPs (53.30 tokens) than GPT-OSS (48.92) and is close to OLMo-3 (51.98). The category trend is consistent: Saul-LM generates longer LPs for *recent* cases (59.84) than for *well-established* cases (46.76). Notably, Saul-LM shows higher variance in LP length, especially for *recent* cases, which suggests less consistent compression behavior than GPT-OSS or OLMo-3.

E.1 Expert Annotation

Figure 2 displays the score distribution of experts' annotation

F Qualitative Evaluation

The two law professors conducted a qualitative examination of the Legal Propositions and of the Expert Evaluation focusing on areas of disagreement, who identified two areas of particular significance:

As discussed above, all five LPs that invalid failed on the *Object* component, i.e. according to one evaluator they failed to express a legal norm. A close re-examination reveal that some of these LPs actually do express a legal norm. For example:

> *An agreement or conduct may be found to have an object of preventing, restricting, or distorting competition under Article 101(1) TFEU even if the parties did not have a subjective intention to do so and pursued legitimate objectives.*

However, LPs sometimes target the Court's own behavior, which might have confused the evaluators. For example:

> *The Court of Justice must take into account the factual and legislative context, as presented in the order for reference, when interpreting EU law under the division of jurisdiction between the Community judicature and national courts.*

Finally, we find that there are LPs that are neither statements of fact nor clearly express a rule or principle. These LPs were not contemplated in the instructions but do provide information about the law that can be used to determine what the law 'is', and as such are valuable. For example:

> *Directive 96/71 was adopted, as indicated by recital 6, to lay down the terms and conditions governing the employment relationship when an undertaking established in one Member State posts workers temporarily to another Member State for the purpose of providing a service, in the interests of both employers and their personnel.*

Based on these findings we conclude that the *Object* criteria generally works well, but that it should be more clear and precise as well as broader to cover statements of the last type.

Table 6 reports a significance difference in quality between well established and recent cases and that the largest gap is found on *source independence*, i.e. that LPs generated from recent cases to a greater extent than well established cases tend to contain references to legal sources. It is not entirely clear why this dimension would be most affected if, as hypothesized, quality difference is driven by the LLMs' pretraining data. A qualitative examination and comparison reveal that LPs generated from well established decisions to a greater extent concern the interpretation of primary law (the Treaties and the Charter), which are more general in nature. Recent decisions, on the other hand, to a greater extent concern secondary law (regulations, directives, or decisions), and normally specific sub- or sub-sub-sections of these instruments. LPs derived from the latter would make little sense without those source references (see for example the LP quoted above as well as those in **??**).

We draw two conclusions from this finding. First, the higher quality of LPs derived from well established cases may in part be driven by LLM pretraining data, but it is likely also in part driven by an underlying difference in characteristics between well established and recent cases. Simply put, one reason why cases become well established is that they concern topics that also make it easier to generate high-quality propositions from them. Related, there is a strong overlap between a case being well established and being frequently cited. Thus, the LLM will have more interpretative context to draw on when generating LPs from well established cases. Second, it causes us to reconsider how source independence is implemented as a quality dimension: a high-quality LP *should not* contain any unnecessary references to specific legal source, but it *should* contain a source reference when this is necessary to make sense of the proposition.